

SAMPLE — B.Tech Computer Science & Engineering

Semester 5 Syllabus (Illustrative / Test Data)

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CS501 - Design and Analysis of Algorithms

Covers asymptotic notation, divide and conquer, dynamic programming, greedy algorithms, graph algorithms, and NP-completeness. Credits: 4. Internal assessment: 30 marks. End-semester exam: 70 marks.

CS502 - Database Management Systems

Covers relational model, SQL, normalization, transactions, concurrency control, and indexing. Credits: 4. Internal assessment: 30 marks. End-semester exam: 70 marks. Lab component: 2 credits.

CS503 - Computer Networks

Covers OSI and TCP/IP models, routing algorithms, congestion control, and application layer protocols. Credits: 3. Internal assessment: 30 marks. End-semester exam: 70 marks.

CS504 - Operating Systems

Covers process management, CPU scheduling, memory management, deadlocks, and file systems. Credits: 4. Internal assessment: 30 marks. End-semester exam: 70 marks.

CS505 - Software Engineering

Covers SDLC models, requirements engineering, software design principles, testing strategies, and project management. Credits: 3. Internal assessment: 30 marks. End-semester exam: 70 marks.

Minimum attendance required: 75% per subject to be eligible to sit for the end-semester examination.